

NELSON, BC, Canada

WMO: 717760

Lat: 49.491N

Lon: 117.305W

Elev: 1755

StdP: 13.79

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.3	15.9	2.9	6.8	14.6	8.0	8.8	18.6	16.5	28.3	14.6	29.4	6.9	250	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	90.1	65.4	86.4	64.4	82.7	63.5	68.1	83.0	66.4	81.1	64.7	79.0	7.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	93.6	73.0	61.5	87.1	70.7	59.7	81.7	69.0	33.5	83.2	32.0	81.3	30.7	79.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.2	11.6	DB	7.2	95.9	5.5	2.8	3.2	98.0	0.0	99.6	-3.1	101.2	-7.1	103.3	(4)
(5)			WB	6.2	71.2	5.3	2.2	2.4	72.8	-0.7	74.1	-3.7	75.3	-7.6	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	30.7	33.0	39.5	47.0	55.2	60.9	68.8	68.3	59.1	46.8	37.4	30.1		
	DBStd	14.76	6.51	5.97	5.35	4.96	5.83	5.49	5.50	5.23	5.70	5.44	5.62	6.40		
	HDD50	2664	599	477	326	116	19	1	0	0	4	128	378	616		
	HDD65	6463	1064	897	790	539	306	147	27	29	191	564	828	1081		
	CDD50	1993	0	0	1	27	181	327	584	567	277	29	0	0		
	CDD65	317	0	0	0	0	3	23	146	130	15	0	0	0		
	CDH74	3778	0	0	0	4	124	381	1666	1406	197	0	0	0		
	CDH80	1375	0	0	0	0	20	108	688	518	41	0	0	0		
	WSAvg	5.1	5.8	5.4	5.3	5.2	5.0	5.1	5.0	4.7	4.5	4.9	5.1	5.5		
	PrecAvg	28.5	3.5	2.0	2.2	1.9	2.4	2.7	1.5	1.3	1.7	2.6	3.2	3.7		
(15) Precipitation	PrecMax	36.1	5.7	4.1	4.4	3.1	5.1	5.1	3.4	3.5	5.2	4.0	6.2	6.0		
	PrecMin	22.8	0.9	0.4	0.6	0.8	0.6	1.5	0.0	0.2	0.1	0.2	1.1	1.9		
	PrecStd	4.3	1.0	0.9	1.1	0.7	1.1	0.8	0.9	1.0	1.2	1.0	1.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	48.5	61.1	72.0	82.7	88.6	95.6	93.3	84.7	67.4	51.7	43.2		
		MCWB	38.9	41.8	47.8	54.0	60.6	64.5	66.3	65.7	63.7	55.8	47.7	41.2		
	2%	DB	39.8	44.4	55.1	66.3	77.4	82.8	91.3	89.5	79.3	62.4	48.9	40.2		
		MCWB	37.8	39.3	44.7	51.5	58.2	62.5	65.7	65.5	61.5	53.3	46.5	38.5		
	5%	DB	38.4	41.7	51.2	62.3	72.9	78.2	87.5	85.8	75.1	58.6	46.6	38.4		
		MCWB	36.9	38.2	42.8	49.3	56.2	60.6	65.1	64.2	60.1	51.4	44.5	37.1		
	10%	DB	37.2	39.6	47.6	58.1	68.5	73.6	83.5	82.2	70.9	55.3	44.7	37.0		
		MCWB	36.0	37.2	41.4	47.2	54.2	59.2	64.3	63.3	58.2	49.9	42.9	35.8		
	0.4%	WB	40.1	43.4	48.9	55.8	62.1	67.3	71.2	70.6	65.8	58.4	49.7	41.9		
		MCDB	40.9	46.3	57.9	69.0	77.2	83.4	86.8	85.3	78.7	64.0	50.4	42.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.4	40.7	46.0	52.7	59.4	64.3	68.7	67.8	63.2	55.7	46.9	39.0		
		MCDB	39.2	42.7	52.7	63.3	73.1	77.9	83.4	82.7	76.0	60.4	48.0	39.7		
	5%	WB	37.2	39.1	44.1	50.3	57.6	62.3	66.9	66.1	61.3	52.9	44.9	37.4		
		MCDB	38.1	40.9	49.4	60.0	69.9	74.4	81.9	80.4	72.4	57.0	46.2	38.1		
	10%	WB	36.2	37.6	42.2	48.3	55.6	60.3	65.2	64.6	59.5	50.5	43.0	36.1		
		MCDB	37.0	39.1	46.6	56.0	66.1	70.8	79.9	78.8	69.2	54.4	44.5	36.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	11.4	15.0	19.8	22.6	21.8	26.7	26.3	22.0	15.0	8.2	6.3		
		MCDBR	7.1	13.8	21.8	28.3	32.1	31.3	33.9	32.5	28.9	20.3	10.5	6.9		
		MCWBR	5.6	9.4	12.7	14.6	14.3	13.3	12.9	13.3	14.3	12.1	7.6	5.7		
		MCDBR	6.6	11.9	19.2	25.2	27.7	27.4	29.6	28.7	26.6	17.4	8.4	6.3		
	5% WB	MCWBR	5.4	8.5	11.8	13.8	13.2	12.6	12.6	13.2	13.9	11.0	6.5	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.284	0.300	0.357	0.384	0.379	0.371	0.386	0.348	0.323	0.298	0.272		
	taud		2.276	2.283	2.293	2.251	2.316	2.375	2.411	2.371	2.471	2.517	2.432	2.298		
	Ebn at Noon		244	274	289	281	278	279	280	270	270	259	238	229		
	Edh at Noon		23	29	34	39	39	37	35	35	29	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		298	590	866	1347	1642	1729	2082	1748	1220	697	313	234		
	RadStd		40	72	129	127	141	160	172	146	136	98	48	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+2.59	N/A	N/A	N/A	N/A	+203	+77
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	N/A

Nomenclature: See separate page